HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
AND ENGINEERING FACULTY OF ECONOMICS

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HCMUTE

SUBJECT: E - COMMERCE SECURITY

Improved fake review detection on Amazon Labeled Fake Reviews dataset using hybrid BERT, behavioral features, CNN-BiLSTM-Attention, Focal Loss, and SHAP explainability

Guidance teacher: Ph.D. Nguyễn Phan Anh Huy

Group: 3

Students perform:

1. Nguyễn Thị Bình Nhi 23126110
2. Đoàn Thị Mỹ Dung 23126068
3. Trương Uyển Duy 23126071
4. Nguyễn Thanh Hảo 23126076
5. Hứa Yến Vy 23126159

Ho Chi Minh City, April, 2026

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Tiêu chí	Yếu	Trung bình	Khá	Giỏi	Điểm GVHD chấm
Thái độ	**0 - 0,4**	**0,5 - 1,0**	**1,1 - 1,6**	**1,7 - 2,0**	
	Rất ít phối hợp nhóm	Thỉnh thoảng phối hợp nhóm	Thường xuyên phối hợp nhóm	Rất thường xuyên phối hợp nhóm	
Hình thức trình bày	**0 – 0,2**	**0,3 - 0,5**	**0,6 – 0,8**	**0,9 – 1,0**	
	Không theo đúng hướng dẫn của môn học (cấu trúc các chương, đánh số các đề mục, font chữ, cỡ chữ, giãn dòng…)	Theo hướng dẫn của môn học , nhưng còn lỗi trong trình bày văn bản, chưa đánh số biểu bảng, đồ thị	Theo hướng dẫn của môn học , còn một số lỗi chính tả và văn phong.	Theo hướng dẫn của môn học (không có lỗi chính tả trong văn bản, hình ảnh bảng biểu rõ ràng, văn phong trong sáng, không có câu tối nghĩa…)	
Phần mở đầu	**0 – 0,2**	**0,3 - 0,5**	**0,6 – 0,8**	**0,9 – 1,0**	
	Không liên quan đến nội dung bài báo cáo.	Nêu được lý do chọn đề tài nhưng chưa trình bày được một số nội dung như: mục tiêu, phạm vi và phương pháp nghiên cứu.	Nêu được đầy đủ các nội dung theo yêu cầu nhưng chưa thực sự thuyết phục.	Nêu được trọn vẹn các nội dung theo yêu cầu, phân tích có tính thuyết phục	
	0 – 1,5	**1,6 - 3,0**	**3,1 – 4,5**	**4,6 – 5,5**	

Nội dung chính	Chỉ giới thiệu thông tin cơ bản về project (địa điểm, quá trình hình thành phát triển) Chưa mô tả được thực trạng của vấn đề nghiên cứu. Chưa có sự liên kết giữa các chương.	Giới thiệu về project nhưng chưa đầy đủ các nội dung theo yêu cầu. Mô tả được thực trạng nhưng chưa đầy đủ, thông tin chưa cập nhật.	Mô tả trung thực, đầy đủ, nhưng còn một số nội dung chưa chi tiết. Demo ít hơn 2 kỹ thuật	Mô tả thực trạng vấn đề nghiên cứu một cách trung thực, đầy đủ, logic, và chi tiết. Nêu được những mặt mạnh, yếu của vấn đề nghiên cứu, demo tốt. Demo nhiều hơn 2 kỹ thuật hoặc demo 1 kỹ thuật phức tạp. Các phân tích, lập luận logic, phù hợp với thực trạng và mục tiêu nghiên cứu.	
Phần kết luận	**0**	**0,1 – 0,2**	**0,3 - 0,4**	**0,5**	
	Kết luận không liên quan đến nội dung báo cáo.	Kết luận chưa tổng quát hóa được vấn đề nghiên cứu.	Kết luận nêu được những điểm nổi bật của báo cáo nhưng chưa chi tiết.	Nêu tóm tắt những điểm nổi bật của báo cáo, nêu những gì đã tìm hiểu, học hỏi được trong quá trình làm project và nêu được hướng nghiên cứu tiếp theo.	
Tổng điểm:					

ASSIGNMENT OF ESSAY WRITING TASKS

SEMESTER II ACADEMIC, 2025 – 2026

Group 3

Subject: E-Commerce Security

Project title: Improved fake review detection on Amazon Labeled Fake Reviews dataset using hybrid BERT, behavioral features, CNN-BiLSTM-Attention, Focal Loss, and SHAP explainability

Num	Full name of Student	Student ID	Completion Rate
1	Đoàn Thị Mỹ Dung	23126068	100%
2	Trương Uyển Duy	23126071	100%
3	Nguyễn Thanh Hảo	23126076	100%
4	**Nguyễn Thị Bình Nhi**	23126110	100%
5	Hứa Yến Vy	23126159	100%

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
AUC	Area Under the Curve
BCE	Binary Cross Entropy
BERT	Bidirectional Encoder Representations from Transformers
Bi-LSTM	Bidirectional Long Short-Term Memory
CFG	Context-Free Grammar
CLIP	Contrastive Language-Image Pre-training
CNN	Convolutional Neural Network
DeBERTa	Decoding-enhanced BERT with disentangled attention
ELECTRA	Efficiently Learning an Encoder that Classifies Token Replacements Accurately
ELMo	Embeddings from Language Models
EU	European Union
FAIR	Facebook AI Research
FGSM	Fast Gradient Sign Method
FTC	Federal Trade Commission

GPT	Generative Pre-trained Transformer
GPU	Graphics Processing Unit
ICCV	International Conference on Computer Vision
ICLR	International Conference on Learning Representations
KDD	Knowledge Discovery and Data Mining
LLM	Large Language Model
LSTM	Long Short-Term Memory
mBERT	Multilingual BERT
MIP	Meaningful Information Patterns
ML	Machine Learning
MLM	Masked Language Modeling
NAACL	North American Chapter of the Association for Computational Linguistics
NBER	National Bureau of Economic Research
NLP	Natural Language Processing
NSP	Next Sentence Prediction
ONNX	Open Neural Network Exchange
PSO	Particle Swarm Optimization

RESTful	Representational State Transfer
ROC	Receiver Operating Characteristic
RoBERTa	Robustly Optimized BERT Approach
SHAP	SHapley Additive exPlanations
SVM	Support Vector Machine
TF-IDF	Term Frequency-Inverse Document Frequency
TPE	Tree-structured Parzen Estimator
ViT	Vision Transformer
WEF	World Economic Forum
WF-CFRB	Weighted Fusion of Contextual Features and Reviewer Behaviors
XAI	Explainable Artificial Intelligence
XLM	Cross-lingual Language Model

GLOSSARY OF MATHEMATICAL SYMBOLS

Symbol	**Meaning**
p_t	Predicted probability of the true class
γ	Focusing parameter in Focal Loss (controls emphasis on hard samples)
α_t	Class balancing weight in Focal Loss
ε	Perturbation magnitude in FGSM adversarial attack
x_adv	Adversarial example generated from original input x
$\nabla_x J$	Gradient of the loss function J with respect to input x
sign(·)	Sign function (returns +1, 0, or -1)
L_total	Combined loss function (natural + adversarial)
α_adv	Weight coefficient for adversarial loss component
L_natural	Loss on original (non-perturbed) samples
L_adversarial	Loss on adversarial samples
$FL(p_t)$	Focal Loss function
$BCE(p_t)$	Binary Cross Entropy loss

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ABSTRACT

In the era of rapid e-commerce expansion, online reviews have become a pivotal factor in determining corporate reputation and consumer purchasing behavior. However, the proliferation of Fake Reviews poses severe threats, undermining market transparency and consumer trust. This research proposes an advanced technical solution leveraging the Amazon Labeled Fake Reviews dataset. The system focuses on constructing a multi-feature classification model that integrates deep semantic insights from BERT embeddings (768 dimensions) with five distinct user behavioral features, resulting in a 773-dimensional composite input vector.

The core architecture utilizes a Hybrid Deep Learning framework combining CNN, Bi-LSTM, and Attention mechanisms, enabling the simultaneous extraction of local features, long-term dependencies, and key linguistic focal points. To enhance performance in real-world scenarios, the study employs Focal Loss to address class imbalance, utilizes the Optuna algorithm for automated hyperparameter optimization, and integrates Fast Gradient Sign Method (FGSM) adversarial training to bolster model robustness against sophisticated attacks. Furthermore, the application of Explainable AI (SHAP) ensures decision-making transparency by providing clear insights into the factors identifying fraudulent reviews.

Experimental results demonstrate that the proposed model outperforms traditional methods and baseline architectures, achieving an Accuracy of 85.70%, a Macro F1-score of 85.46%, and a ROC-AUC of 91.33%. Notably, the model achieves a precision of 96.53% for the "Fake" class, significantly reducing the risk of false positives against legitimate customer feedback. This solution not only advances fake review detection but also safeguards consumer trust, mitigates system manipulation risks, and supports e-commerce platforms in complying with AI regulations, such as the EU AI Act (2024).

1. INTRODUCTION

1.1. Rationale for the Research

The exponential growth of e-commerce has transformed online review systems into a decisive element for consumer behavior and brand equity. According to recent reports, 95% of consumers consult reviews before purchasing, influencing 32% of total transaction volumes on digital platforms (Shapo AI Shopping Report, 2026).

From an E-commerce Security perspective, fake reviews represent a deliberate form of Information Manipulation attack. Malicious actors utilize botnets and Generative AI to produce fraudulent reviews with natural language and authentic emotional tones, easily bypassing traditional filters. It is estimated that fake reviews caused approximately $770.7 billion in global consumer losses in 2025, a figure projected to reach $1.07 trillion by 2030 (Capital One Shopping Research, 2026). The World Economic Forum (WEF) also notes direct costs to businesses amounting to $152 billion annually. This represents a critical security vulnerability, allowing fabricated data to interfere with recommendation algorithms and product rankings.

A major challenge lies in the fact that fake reviews are growing 12.1% faster than authentic ones, coupled with the rise of "AI slop"—spam content generated by AI, which is increasing by 80% per month (ReviewDriver, 2025). Currently, between 30% and 43% of reviews on major e-commerce platforms are suspected of being fraudulent.

In response, this research proposes a solution based on a Hybrid Deep Learning model combining BERT embeddings and user behavioral features. The system integrates Focal Loss for imbalanced data handling and FGSM Adversarial Training to enhance resilience against adversarial attacks. The integration of SHAP improves the transparency and explainability of the model. This study contributes to the advancement of Natural

Language Processing (NLP) techniques and addresses urgent security requirements in e-commerce to protect consumer trust and digital economy sustainability.

1.2. Research Objectives

The primary objective is to develop a Hybrid Deep Learning model that integrates semantic and behavioral features to detect fake reviews on e-commerce platforms with high accuracy and robustness. Specifically:

Feature Engineering: Extract semantic features using BERT and integrate them with five behavioral metrics to create a multi-dimensional input vector.

Class Imbalance Mitigation: Apply Focal Loss to handle the data distribution disparity between genuine and fraudulent reviews.

Optimization: Utilize Optuna for automated hyperparameter tuning.

Interpretability: Implement Explainable AI (SHAP) to clarify decision-making mechanisms, enhancing system transparency and auditability.

Robustness Testing: Evaluate model resilience against adversarial attacks, specifically focusing on AI-generated fake content.

1.3. Research Content

The study encompasses the following key areas:

Theoretical research on BERT architecture, CNN, Bi-LSTM, Attention mechanisms, and optimization techniques (Focal Loss, FGSM, SHAP).

Data preprocessing of the Amazon Labeled Fake Reviews dataset, including semantic and behavioral feature extraction. Design and implementation of the Hybrid Deep Learning architecture.

Empirical evaluation and benchmarking against baseline models using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Interpretability analysis via SHAP to determine the impact of specific feature groups on classification.

1.4. Scope of the Research

Object of Study: Textual reviews and associated user behavioral data on e-commerce platforms.

Experimental Dataset: Amazon Labeled Fake Reviews dataset.

Technical Limits: Focused on modern NLP techniques (Transformers/BERT) and Deep Learning; hardware infrastructure implementation is outside the scope.

1.5. Research Methodology

Data Collection: Utilizing a standardized, labeled dataset for fake review detection.

Analysis and Synthesis: Reviewing scientific literature and international journals on NLP and fraud detection to build a theoretical framework.

Quantitative Experimentation: Developing an automated data pipeline using Python (PyTorch/TensorFlow, HuggingFace, and Optuna) and conducting Adversarial Training to test model durability.

Evaluation: Comparing experimental results through convergence plots and standard machine learning performance metrics.

2. LITERATURE REVIEW

2.1. The Challenge of Fake Reviews in E-commerce

The rapid growth of e-commerce over the past decade has fundamentally changed how consumers access information before making purchasing decisions. Instead of experiencing products directly as in traditional stores, online shoppers rely heavily on reviews from previous customers. Data from the BrightLocal Consumer Review Survey (2026) shows that approximately 97% of consumers regularly read online reviews before making a purchase decision, and on average, they need to read at least 10 reviews to build trust with a new brand. This reality places online review systems as one of the key factors influencing shopping behavior in the digital environment (Lim et al., 2025).

However, the importance of online reviews also creates fertile ground for fraudulent reviews to flourish. Fake reviews, also known as *opinion spamming,* are defined as comments created without actual experience with a product or service, with the aim of manipulating consumer perceptions or harming competitors (Salminen et al., 2022). These reviews can be falsely positive to inflate product reputation, or intentionally negative to discredit competitors.

The scale of the fake review problem has reached alarming levels. Hollenbeck's (2025) study of nearly 1,500 Amazon products found that, on average, about 47% of 5-star reviews on products with fake reviews are inauthentic. Backlinko (2026) noted that 44% of consumers believe they have encountered fake reviews on Amazon, and 40% suspect the same on Google. On a platform scale, TripAdvisor blocked over 2 million fake reviews in 2023, while Trustpilot removed nearly 3.8 million invalid reviews in 2024 (Shapo, 2025).

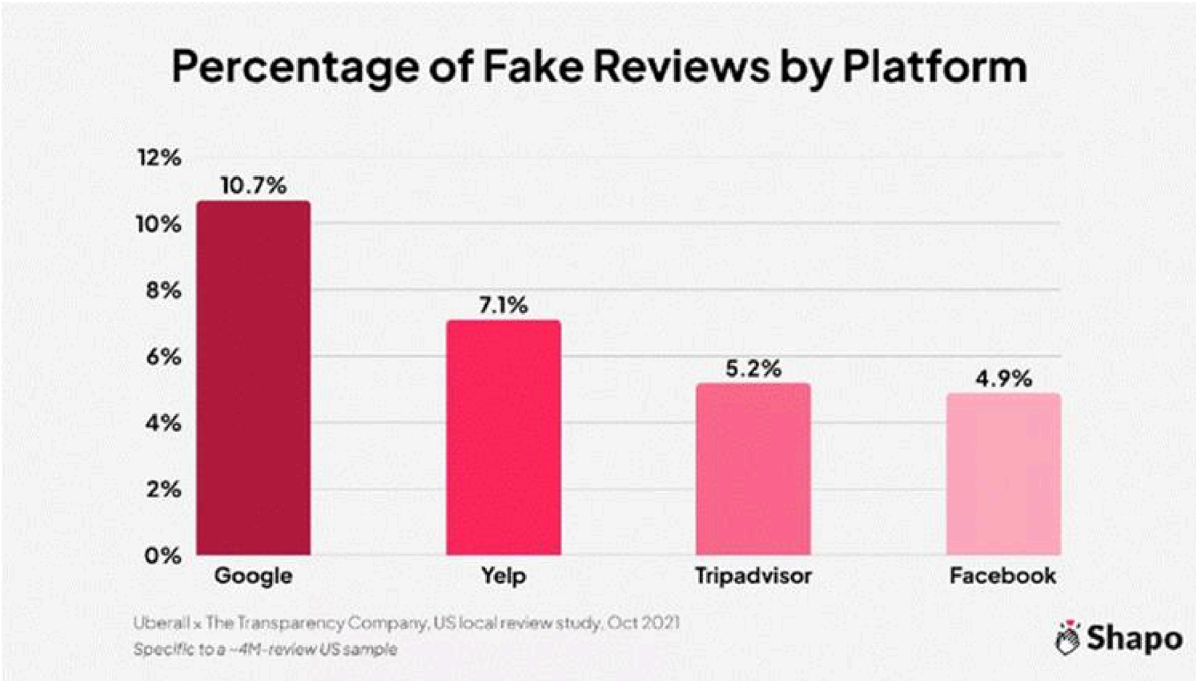

Figure 2.1. Fraudulent rating statistics by platform

(Source: Shapo, 2025)

The bar chart compares the percentage of fake reviews on Google (10.7%), Yelp (7.1%), Tripadvisor (5.2%), and Facebook (4.9%). This visually illustrates the different scales of the fake review problem across platforms, highlighting that Amazon and Google are the most heavily affected, thus clarifying the reason for focusing the study on the Amazon dataset.

The challenges of detecting false reviews relate to the following issues:

Firstly, there is a data imbalance: in the Amazon Labeled Fake Reviews dataset used in this study, genuine reviews accounted for approximately 81% while fake reviews accounted for only 19%, a typical discrepancy in real-world data.

Secondly, there is the issue of linguistic diversity: fake reviews can be written very cleverly, mimicking natural writing styles and difficult to distinguish with the naked eye.

Thirdly, there is the continuous evolution of cheating tactics, especially in the context of large language models like ChatGPT or GPT-4 which can generate very realistic-sounding fake reviews. Salminen et al. (2022) demonstrated that reviews generated by GPT-2 can fool most ordinary readers.

Fourth is the need for transparency and accountability: platforms not only need to know which reviews are fake but also need specific reasons to treat them fairly and comply with increasingly stringent AI regulations.

2.2. The impact and consequences of fake reviews

The consequences of fake reviews extend beyond consumers buying substandard products; they also have far-reaching negative impacts on the entire e-commerce ecosystem.

Economically, Akesson et al. (2023) in a study by the U.S. National Bureau of Economic Research (NBER) demonstrated that fake reviews lead consumers to choose lower-quality products, resulting in welfare losses of up to $0.12 for every dollar spent. On a larger scale, the Federal Trade Commission (FTC, 2022) estimates billions of dollars in losses annually for American consumers due to erroneous purchasing decisions.

For legitimate businesses, fake reviews from competitors create an unfair competitive environment. Hollenbeck's (2025) research shows that products with artificially inflated reviews can maintain higher average ratings, attracting more purchases despite lower actual quality. This creates a negative cycle: honest businesses suffer revenue and market share losses, the market is distorted, and ultimately consumers bear the consequences.

Regarding consumer trust, BrightLocal (2026) noted a significant increase in the percentage of users who “always” read reviews before searching for information about businesses compared to the previous year. More than 85% of consumers are more likely to use a service after reading positive reviews, while 77% will abandon their intention to buy when encountering negative reviews. This shows that fraudsters can reap huge profits from manipulating review systems.

From an e-commerce security perspective, fake reviews are a direct attack on the integrity and trustworthiness of the review system – a core component of e-commerce platforms. Successful manipulation of this system not only erodes user trust but also allows interference with ranking algorithms, recommendation systems, and even influences the platform's business decisions.

From a legal and governance perspective, in 2023, Amazon, Booking.com, Google, TripAdvisor, and several other major platforms established the “Coalition for Trusted Reviews” to coordinate the development of technology to combat fake reviews (BrightLocal, 2026). In the European Union, the Omnibus Directive 2019/2161/EU requires platforms to verify the source of reviews, while the AI Act 2024 mandates that high-risk AI systems must ensure explainability and accountability – setting mandatory requirements for XAI (European Commission, 2024).

A prime example is Amazon's 2025 lawsuit against a network of over 75 websites that provided services selling fake reviews and fake seller accounts. The creation of a sophisticated, organized underground industry for fake reviews demonstrates the severity of this security threat. Platforms spend hundreds of millions of dollars annually on both legal measures and detection technology.

These impacts confirm that fake reviews are not just a content quality issue, but also a real security threat to data integrity, user trust, and the stability of the e-commerce ecosystem.

2.3. Related studies in the field of spurious evaluation detection

Research on spurious evaluation detection has gone through three distinct stages of development: from traditional machine learning methods, to deep learning, and most recently, exploiting large language models based on the Transformer architecture.

Phase 1: Traditional Machine Learning Methods

Pioneering studies, notably Ott et al. (2011) and Mukherjee et al. (2012), primarily used linguistic features (n-grams, TF-IDF) in combination with classifiers such as Naive Bayes, SVM, or Random Forest. Mukherjee et al. (2012) proposed a method for detecting fake reviewer groups on Yelp based on behavioral and linguistic features. However, the fundamental drawback is its reliance on handcrafted features *and* its inability to capture the deeper semantics of the text.

Phase 2: Deep Learning and Neural Networks

The emergence of CNN, LSTM, and BiLSTM architectures has opened up new approaches. Alsubari et al. (2021) proposed an integrated CNN-LSTM model for pseudo-reviewer classification across multiple domains, achieving 87% accuracy on the Amazon dataset. Deshai and Bhaskara Rao (2023) combined CNN with an adaptive swarm optimization (PSO) algorithm, showing that combining deep learning with optimization algorithms significantly improves performance. Zhang et al. (2023) exploited reviewer behavioral features, demonstrating the added value of this behavioral information directly inherited in the pipeline design with the five behavioral features of this study.

Phase 3: Transformers and BERT (2019–present)

The emergence of BERT (Devlin et al., 2019) marked a turning point in natural language processing. Salminen et al. (2022) fine-tuned RoBERTa to create FakeRoBERTa, achieving superior results. Mohawesh et al. (2024) proposed a combined Transformer-BiLSTM-RoBERTa architecture. Sun et al. (2024) published a model combining BERT with reviewer and merchant behavioral features on the Dianping platform, demonstrating the trend of integrating semantic and behavioral information as a dominant direction. Wang and Chen (2025) proposed WF-CFRB (Weighted Fusion of Contextual Features and Reviewer Behaviors), a latest architecture (published March 2025 in *the Journal of Systems Science and Systems Engineering*) that combines BERT for semantic features and CNN for behavioral features through a weighted fusion mechanism, achieving superior performance across multiple domains. This design is the closest in philosophy to the pipeline of this study.

Gupta et al. (2024) summarized the state of the field and concluded that deep learning models outperform traditional machine learning on large datasets, but *hybrid models* combining multiple components often yield the best results. The survey also pointed out three research gaps: a lack of systematic data imbalance handling, a lack of techniques for strengthening resilience against adversarial attacks, and a lack of transparency in decision interpretation – these are the three issues that the pipeline in this study addresses through Focal Loss, FGSM, and SHAP.

Table 2.1. Summary of related studies in the field of spurious assessment detection.

Author & Year	**Dataset**	**Method**	**Best results**	**Main limitations**	**Stage**
Ott et al. (2011)	TripAdvisor (hotels)	SVM + n-gram, POS tags, CFG parse	Accuracy 89.8%	Only linguistic features are considered; imbalances are not addressed.	Traditional ML
Mukherjee et al. (2012)	Yelp (restaurant)	Behavioral clustering + Naïve Bayes	Precision ~80%	Relys on manual processes; difficult to	Traditional ML

				expand to new domains.	
Alsubari et al. (2021)	Amazon, Yelp, IMDb (multi-domain)	CNN-LSTM	Accuracy 87% (Amazon)	No imbalance correction; no bidirectional or attention control.	Deep learning
Deshai & Bhaskara Rao (2023)	Ott, Amazon, Yelp, TripAdvisor, IMDb	CNN + Adaptive PSO	F1 is an improvement over pure CNN.	PSO is resource-intensive; do not use pre-trained LM.	Deep learning
Salminen et al. (2022)	Amazon (multi-category)	Fine-tuned RoBERTa (FakeRoBERTa)	Superior to baseline	No integration of behavioral features; no explanation of the decision.	Transformer

Sun et al. (2024)	Dianping (multidisciplinary)	BERT + Transformer + reviewer/merchant behavior	AUC and F1 improved compared to BERT alone.	The dataset is primarily in Chinese; multilingual support has not yet been tested.	Transformer
Wang & Chen (2025)	Amazon, Yelp	WF-CFRB: BERT + CNN (weighted fusion context + behavior)	Superior baseline performance across multiple domains.	High computational cost; requires a large amount of behavioral data.	Transformer
Mohawesh et al. (2024)	Multiple domains	Transformer + BiLSTM + RoBERTa	State of the Art across multiple issues.	Complex, high computational cost; lacks XAI.	Transformer
This study	**Amazon Labeled Fake**	**BERT + CNN-BiLSTM-Attention + Focal Loss**	**F1 macro optimized via Optuna**	**Currently under review**	**Transformer + Hybrid**

	Reviews (~50,000 lines)	**+ Optuna + FGSM + SHAP**		**— see Chapter 3**	

(Source: Compiled by the research team)

3. METHOD

3.1. Text processing and classification techniques (BERT and Large Language Models)

One of the core challenges in the spurious evaluation detection problem is how to make the computer "understand" the text with sufficient semantic depth. Text representation methods have evolved from bag-of-words and TF-IDF, which do not capture context, to large-scale language models based on attention mechanisms.

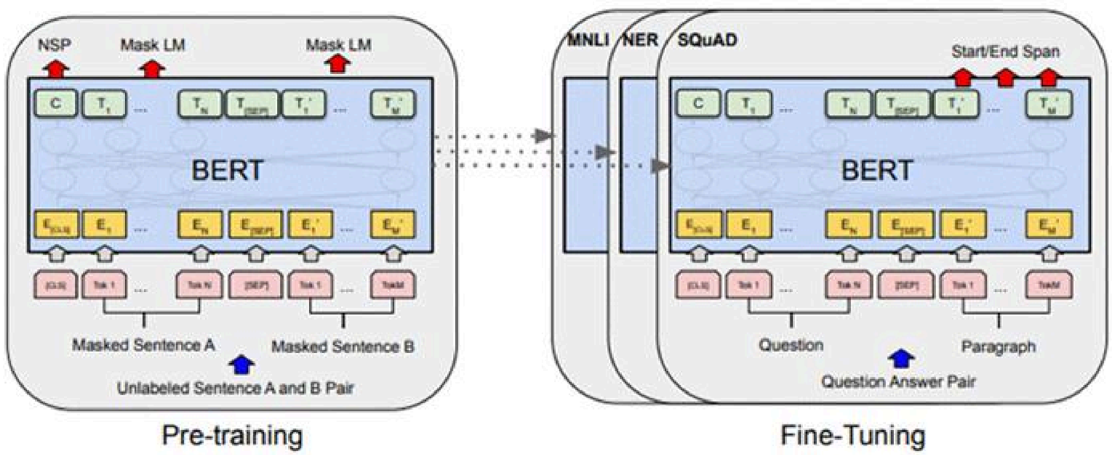

Figure 3.1. BERT-base-uncase architecture

(Source: Devlin et al. (2019))

BERT (Bidirectional Encoder Representations from Transformers), introduced by Devlin et al. (2019) at the NAACL conference, is a breakthrough in natural language processing. The core difference between BERT and previous models is its ability to encode context bidirectionally: during training, BERT simultaneously considers the left and right contexts of each word, instead of only one direction like GPT or ELMo. BERT is pre-trained on a massive corpus using two tasks: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). The *bert-based-uncased version* has 12 Transformer classes, 12 attention heads, a hidden size of 768, and 110 million parameters.

A comprehensive survey of BERT applications in NLP by Gardazi et al. (2025) confirms that BERT remains the best platform for text classification tasks in resource-constrained environments.

In this study, BERT was used in a feature extraction method instead of fine-tuning. The entire review text was processed through BERT to obtain a 768-dimensional [CLS] vector, which was then combined with five behavioral feature vectors to create a 773-dimensional input vector for the classification model. The five behavioral features include: (1) rating *detects* reviews with extreme scores (1 or 5 stars); (2) *verified_purchase* detects unverified reviews with a higher likelihood of being fake; (3) *helpful_vote* detects fake reviews that are less likely to be helpful; (4) *user_review_burst* detects unusually rapid review behavior; and (5) *rating_sentiment_gap* = |rating − sentiment_score| detects a discrepancy between score and content. This approach is consistent with the trend described by Sun et al. (2024) and Wang & Chen (2025) confirm that combining semantic and behavioral features yields significantly better results than using only text.

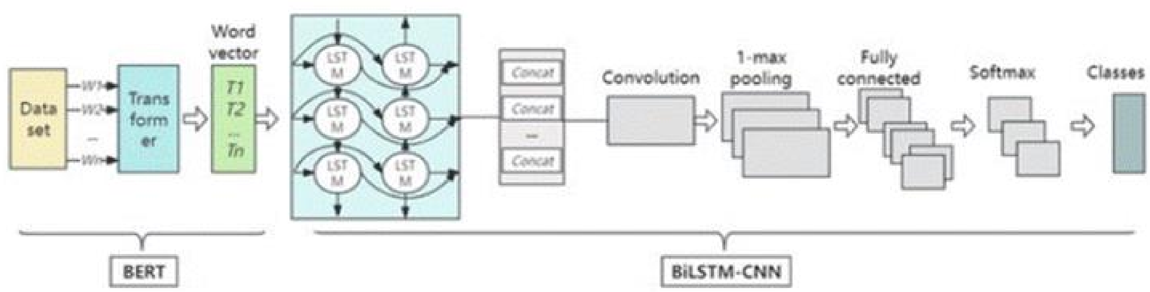

Figure 3.2. Pipeline integrating BERT features (768 dimensions) and behavioral features (5 dimensions) to create a 773-dimensional input vector.

(Source: researchgate, 2024)

Besides BERT, recent studies have explored larger models such as RoBERTa, GPT-3, GPT-4, and DeBERTa. Thao et al. (2024) developed DenyBERT, combining BERT with Knowledge Distillation and DeLighT techniques, requiring only 16.01M parameters but achieving 96.12% accuracy and a 96.47% F1-score, demonstrating that lightweight models can achieve high performance. However, GPT-3 cannot fine-tune large datasets, which is a practical reason why BERT-based models remain a suitable choice for pipelines processing around 50,000 evaluations, as in this study.

3.2. Techniques for handling imbalanced data and optimization

3.2.1. Focal Loss

Focal Loss was proposed by Lin et al. (2017) of Facebook AI Research (FAIR) in their paper *"Focal Loss for Dense Object Detection"* presented at ICCV 2017. Although initially designed for object detection in computer vision, Focal Loss has quickly become widely applied in classification problems with disequilibrium data.

The core idea of Focal Loss stems from the observation that most of the updated gradient comes from "easy" samples—those that the model has already classified correctly with high confidence. Focal Loss addresses this problem by adjusting the Binary Cross-Entropy function through an adjustment factor:

$$\mathbf{FL(p_t) = -\alpha_t \times (1 - p_t)^{\gamma} \times \log(p_t)}$$

In this model, p_t is the probability that the model correctly predicts the label; γ (gamma) is the focal loss parameter (when $\gamma = 0$, Focal Loss is equivalent to a regular BCE); α_t is the class balance parameter. The factor $(1 - p_t)^{\gamma}$ is the key point: when the model classifies correctly with high confidence ($p_t \to 1$), this factor approaches 0 and reduces the sample's contribution to the total loss. Conversely, with a difficult-to-classify sample, the factor is close to 1, maintaining a high loss so the model continues to learn.

In this study, the parameter γ was automatically optimized via Optuna with a search space from 0.5 to 3.0, and $\alpha_t = 0.25$ as recommended by Lin et al. (2017). Optuna determined $\gamma = 2.5$ to be optimal on the validation set.

3.2.2. Optuna — Bayesian Optimization

Optuna, introduced by Akiba et al. (2019) at KDD, is a next-generation hyperparameter optimization framework with a *"define-by-run" philosophy* . Its core algorithm is the TPE Sampler (Tree-structured Parzen Estimator), a Bayesian Optimization variant that builds two distribution models: l(x) for good parameters and g(x) for poor parameters, maximizing the l(x)/g(x) ratio. This method is superior to grid search because it learns from the history of trials to direct the search towards more promising parameter regions.

Table 3.1. Hyperparameter search space in Optuna

hyperparameters	**Search range**	**Purpose**
filters (Conv1D)	64 / 128 / 256	Number of CNN filters that extract n-gram features
kernel_size	2 / 3 / 4	Sliding window sizes (unigram, bigram, trigram)
lstm_units	32 / 64 / 128	Bi-LSTM neurons learn two-way context.
learning_rate	$1\times10^{-5} \rightarrow 5\times10^{-4}$	Learning speed affects convergence speed.
dropout_rate	0.3 / 0.5 / 0.7	Dropout rate against overfitting

batch_size	16 / 32 / 64	Number of samples per weight update step
focal_gamma (γ)	0.5 → 3.0	The level of concentration on pseudo-patterns that are difficult to classify.

(Source: Design by the research team)

In this study, Optuna was configured to maximize the F1-score macro on a validation set across 20 trials. After optimization, the best set of parameters was determined: conv_filters = 128, kernel_size = 2, lstm_units = 64, learning_rate = 0.00034, dropout_rate = 0.3, batch_size = 32, and focal_gamma = 2.5.

3.3. Adversarial Training (FGSM) and Explainable AI (SHAP)

3.3.1. FGSM Adversarial Training

Adversarial examples are inputs generated by intentionally adding small noises to trick a machine learning model. This concept was systematized by Goodfellow et al. (2015) at ICLR 2015. The Fast Gradient Sign Method (FGSM), proposed in the same paper, is the simplest and most efficient technique for generating adversarial examples:

$$\mathbf{x\_adv = x + \varepsilon \times sign(\nabla_x J(\theta, x, y))}$$

Where x is the original input; ε (epsilon) is the noise magnitude; $\mathrm{sign}(\nabla_x J(\theta, x, y))$ is the sign of the gradient of the loss function J with respect to the input x. The combined loss function in adversarial training:

$$\mathbf{L\_total = (1 - \alpha\_adv) \times L\_natural + \alpha\_adv \times L\_adversarial}$$

With $\alpha\_adv = 0.35$ and $\varepsilon = 0.005$, the model was trained to maintain classification performance on both the original data and the data with added noise. In the context of spurious review detection, adversarial training has particular practical significance: it simulates a situation where a fraudster intentionally manipulates the content of a spurious review to deceive the system, such as adding positive words, slightly altering sentence structure, or adjusting scores. This is a direct response to the growing concern about spurious reviews generated by ChatGPT or GPT-4.

3.3.2. SHAP — Explainable AI

SHAP (SHapley Additive explanations) was proposed by Lundberg and Lee (2017) at NeurIPS 2017. SHAP provides a unified framework for explaining predictions of any machine learning model, based on Shapley values from cooperative game theory. Lundberg and Lee (2017) demonstrated that SHAP is the only method that simultaneously satisfies four properties: efficiency, symmetry, dummy, and additivity.

Recent research by Salih et al. (2025) in the journal Advanced Intelligent Systems (Wiley) proposes a more comprehensive SHAP interpretation framework, combined with MIP (Meaningful Information Patterns) techniques to address the correlation problem between features. This is a potential direction for future research. Furthermore, research by Mustafa and Hama Saeed (2025) in the Journal of Electrical Systems and Information Technology confirms that combining SHAP with NLP models significantly increases transparency and user trust in AI systems.

In this study, a model-agnostic variant of SHAP KernelExplainer was used, suitable for the complex CNN-BiLSTM-Attention architecture. KernelExplainer approximates SHAP values by solving a weighted linear regression problem with a special kernel, without requiring internal model information. The SHAP results allow for the identification of features that contribute most to classification decisions.

The transparency that SHAP brings is valuable not only scientifically but also legally and commercially. In the context of the EU AI Act (2024), which requires high-risk AI systems to be able to explain decisions, integrating SHAP is a necessary step to ensure the system meets accountability requirements in practical implementation.

3.4. Overview of the Pipeline Process

In this study, the team proposes a complete pipeline to address the problem of detecting fake reviews on e-commerce platforms. The pipeline is designed to combine both text content and user behavior characteristics, thereby improving the ability to detect sophisticated fake reviews—especially those with natural language that are difficult to distinguish with the naked eye.

The defining characteristic of this pipeline is its multi-layered integration: from data preprocessing, semantic feature extraction (BERT), user behavior feature integration, deep classification model building, to advanced optimization techniques and model interpretation. Each step in the pipeline adds information that the previous step has not yet exploited.

The overall process comprises six main steps, described in the following order:

Step 1 – Data collection and preprocessing

Input data includes review content (title and content), rating, verified purchase status, helpful vote count, and other metadata fields related to user behavior.

Step 2 – Feature Extraction

The text is represented as a numerical vector using the main method:BERT embedding.

Step 3 – Multi-feature Fusion

Semantic features from the text are integrated with five user behavior features to form a multidimensional input vector, adding extralinguistic context to the classification process.

Step 4 – Build the classification model

The team built and compared two deep learning architectures: Hybrid Deep Learning (CNN + BiLSTM + Attention) and Transformer Encoder.

Step 5 – Optimization and Training

The training process is enhanced by three supplementary techniques: Focal Loss to handle data imbalances, Optuna to find optimal hyperparameters using Bayesian Optimization, and FGSM Adversarial Training to increase model robustness.

Step 6 – Model Evaluation

Model performance is measured using a comprehensive set of metrics including Accuracy, F1-score (macro), Precision, Recall, and ROC-AUC, meeting both overall classification criteria and the ability to detect minority classes (fake review).

Table 3.2. Overview of the proposed fake review detection pipeline.

STT	Step name	Detailed description	Technology / Method
1	Collection & Preprocessing	Clean up text, handle missing values, normalize labels.	Pandas, NLTK
2	Feature extraction	Representing text as a numerical vector	BERT
3	Multi-feature combination	Combining BERT embeddings + 5 behavioral features → 773-d	torch.cat()
4	Building a model	Hybrid or Transformer Classification Architecture	PyTorch
5	Optimization	Imbalance handling + hyperparameter optimization + adversarial	Focal Loss, Optuna, FGSM
6	Review & Explanation	Measuring performance and explaining modeling decisions.	SHAP

(Source: Design by the research team)

This pipeline ensures that the model not only learns the linguistic content of reviews but also understands unusual user behavior—a crucial element often overlooked in previous studies.

3.5. Description of the Amazon Labeled Fake Reviews dataset

The team uses the dataset.Amazon Labeled Fake Reviews,It was publicly released on the Kaggle platform a year ago. This is one of the datasets commonly used in research on detecting fake reviews in e-commerce.

Table 3.3 – Overview of the characteristics of the Amazon Labeled Fake Reviews dataset

Characteristic	**Detail**
Data scale	Approximately 50,000+ samples
Number of attributes (columns)	13 columns
Text components	Title and text of the review
Metadata component	Rating, verified purchase, helpful vote, ...
Class 0 label	Genuine (Real) Review
Grade 1 Label	Fake reviews
Label distribution ratio	~81% Real – ~19% Fake (imbalance)

(Source: Design by the research team)

Data components consist of two main groups of attributes:

Group 1 – Text Features: Includes the review title and review text. This is the primary linguistic information source used to extract semantic representations through BERT embedding.

Group 2 – User Behavior Characteristics: Includes columns such as rating, verified_purchase, helpful_vote, and other additional information reflecting the context of submitting the review.

A key and most challenging characteristic of this dataset is the severe imbalance between the two classes: genuine reviews account for approximately 81% while fake reviews only account for about 19%. This causes conventional classification models to tend to favor the majority class (genuine reviews), leading to a low fake review detection rate. To address this issue, the team applied the Focal Loss function instead of the usual Binary Cross Entropy. The mechanism and reasons for this choice will be detailed in Section 2.3.3

3.6. Steps to upgrade the model (Model 2 – The main proposed model)

3.6.1. Multi-feature extraction

As a first step, the team used BERT embedding to significantly improve the model's ability to understand deep semantics.

Specifically, the input text—including both the title and the review text—is concatenated and passed through a BERT-based-uncased model. After tokenization, the input string is processed through the entire BERT Transformer architecture, and the semantic representation vector of the sentence is derived from the [CLS] token in the final layer, creating a 768-dimensional embedding. This representation allows the model to capture contextual semantics.

```python
import numpy as np
from transformers import BertTokenizer, BertModel
import torch
from tqdm import tqdm

tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
model = BertModel.from_pretrained('bert-base-uncased')
model.eval()
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
model = model.to(device)
print("Đang dùng:", device)

def get_bert_embedding(texts, batch_size=32):
    all_embeddings = []
    for i in tqdm(range(0, len(texts), batch_size), desc="BERT đang chạy"):
        batch = list(texts[i:i+batch_size])
        encoded = tokenizer(batch, padding=True, truncation=True,
                            max_length=256, return_tensors='pt').to(device)
        with torch.no_grad():
            output = model(**encoded)
        all_embeddings.append(output.last_hidden_state[:, 0, :].cpu().numpy())
    return np.vstack(all_embeddings)

bert_vectors = get_bert_embedding(df['combined_text'])
print("✅ Shape BERT:", bert_vectors.shape)
```

Figure 3.3. ***The process of extracting BERT embeddings from review documents.***

(Source: Design by the research team)

Simultaneously, the team retained the five user behavior characteristics from the baseline: rating, verified_purchase, helpful_vote, user_review_burst, and rating_sentiment_gap. Notably, the rating_sentiment_gap characteristic plays a crucial role in detecting potential discrepancies between user ratings and the actual emotional tone of the review content—a hallmark of many types of fake reviews.

Table 3.4. Five user behavior characteristics used in the model

Feature	**Meaning**
rating	The rating scale from 1–5 stars is given to products by users.
verified_purchase	Verify if the user actually purchased the product (0/1).
helpful_vote	The number of helpful votes from the community.

user_review_burst	Behavior detection indices assess a range of abnormalities over a short period of time.
rating_sentiment_gap	The discrepancy between the score and the actual emotions expressed in the text is a characteristic sign of a fake review.

(Source: Design by the research team)

```python
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer
from sklearn.preprocessing import StandardScaler
import numpy as np

analyzer = SentimentIntensityAnalyzer()

# rating_sentiment_gap
def sentiment_to_rating(text):
    score = analyzer.polarity_scores(str(text))['compound']
    return (score + 1) / 2 * 4 + 1

df['sentiment_score'] = df['text'].apply(sentiment_to_rating)
df['rating_sentiment_gap'] = abs(df['rating'] - df['sentiment_score'])

# user_review_burst
burst_col = 'reviewer_id' if 'reviewer_id' in df.columns else None
df['user_review_burst'] = df[burst_col].map(df[burst_col].value_counts()) if burst_col else 1

# verified_purchase → 0/1
df['verified_purchase_num'] = df['verified_purchase'].apply(
    lambda x: 1 if str(x).lower() in ['true', '1', 'yes'] else 0)

# helpful_vote → fill 0
df['helpful_vote'] = df['helpful_vote'].fillna(0)

# Chuẩn hóa + ghép 773 chiều
behavior = df[['rating', 'verified_purchase_num', 'helpful_vote',
               'user_review_burst', 'rating_sentiment_gap']].values
behavior_scaled = StandardScaler().fit_transform(behavior)
X = np.concatenate([bert_vectors, behavior_scaled], axis=1)
y = df['label'].values

print("✅ Shape X:", X.shape)  # → (50000, 773)
print("✅ Shape y:", y.shape)

# Lưu lên Drive
np.save('/content/drive/MyDrive/X_features_773.npy', X)
np.save('/content/drive/MyDrive/y_labels.npy', y)
print("✅ Lưu xong lên Drive!")
```

Figure 3.4 Behavioral features combined with BERT create a 773-dimensional vector.

(Source: Design by the research team)

Finally, the 768-dimensional BERT vector is concatenated with 5 behavioral features, creating a combined 773-dimensional input vector. This multi-feature combination allows the model to learn simultaneously from two complementary sources of information: linguistic semantics and user behavior, thereby improving its ability to detect fake reviews in many complex situations.

3.6.2. Hybrid Deep Learning Classification Model (CNN+Bi-LSTM+Attention)

The classification model architecture is designed as a hybrid, leveraging the strengths of three different deep learning components: CNN, BiLSTM, and Attention. Each architecture individually addresses only one aspect of the text. CNN excels at detecting local features (n-grams), LSTM excels at learning long-term sequential relationships, and Attention helps focus on important parts. Combining the three architectures creates a more comprehensive classifier.

Table 3.5. Detailed structure of the Hybrid Deep Learning model

Ingredient	**Configuration**	**Function**
Input layer	773-dimensional vector	Get BERT vector + 5 behavioral features
Conv1D (CNN)	128 filter, kernel size = 3	Extracting local features (n-grams) to detect characteristic phrases of fake reviews.
BiLSTM	64 units (each way)	Learn contextual relationships in two directions (left → right and right → left).
Attention Layer	Self-learning weights	Automatically focus on the most important locations in the chain.
Dropout	Rate = 0.5	Reduce overfitting during training.

Dense + Sigmoid	1 output node	Binary classification: probability is Fake (0→1)

(Source: Design by the research team)

```
class CNN_BiLSTM_Attention(nn.Module):
    def __init__(self, input_dim=773, conv_filters=128, kernel_size=3,
                 lstm_units=64, dropout_rate=0.5):
        super().__init__()

        # Conv1D: reshape input thành (batch, 1, 773)
        self.conv1 = nn.Conv1d(in_channels=1, out_channels=conv_filters,
                               kernel_size=kernel_size, padding=1)
        self.relu = nn.ReLU()
        self.maxpool = nn.MaxPool1d(kernel_size=2, stride=2)

        # Bi-LSTM
        self.lstm = nn.LSTM(input_size=conv_filters,
                            hidden_size=lstm_units,
                            bidirectional=True,
                            batch_first=True)

        # Attention layer (tự học trọng số quan trọng)
        self.attention = nn.Linear(lstm_units * 2, 1)

        # Fully connected
        self.dropout = nn.Dropout(dropout_rate)
        self.fc1 = nn.Linear(lstm_units * 2, 64)
        self.fc2 = nn.Linear(64, 1)
        self.sigmoid = nn.Sigmoid()

    def forward(self, x):
        # x: (batch, 773)
        x = x.unsqueeze(1)                  # (batch, 1, 773)

        # CNN block
        x = self.conv1(x)                   # (batch, 128, ~773)
        x = self.relu(x)
        x = self.maxpool(x)                 # (batch, 128, ~386)

        # Bi-LSTM
        x = x.transpose(1, 2)               # (batch, seq_len, 128) cho LSTM
        lstm_out, _ = self.lstm(x)          # (batch, seq_len, 128)

        # Attention
        attn_scores = self.attention(lstm_out).squeeze(-1)   # (batch, seq_len)
        attn_weights = torch.softmax(attn_scores, dim=1)     # (batch, seq_len)
        context = torch.bmm(attn_weights.unsqueeze(1), lstm_out).squeeze(1)  # (batch, 128)

        # Dense layers
        x = self.dropout(context)
        x = self.fc1(x)
        x = self.relu(x)
        x = self.fc2(x)
        x = self.sigmoid(x)

        return x
```

Figure 3.5 Define the CNN_BILSTM_Attention model.

(Source: Design by the research team)

The data flow in the model is described as follows: the 773-dimensional input vector passes through the Conv1D layer to extract local features, then through BiLSTM to learn two-dimensional context, the Attention mechanism assigns weights to each position, then through Dropout for regularization, and finally through Dense + Sigmoid to produce classification probabilities.

3.6.3. Handling Data Imbalances Using Focal Loss

Due to the significant imbalance between the two classes in the dataset (81% genuine reviews versus 19% fake reviews), the team used Focal Loss. Focal Loss was initially proposed by Lin et al. (2017) for the problem of object detection in images and has been shown to be effective in class imbalance situations.

Focal Loss Formula:

$$\mathbf{FL(p_t) = -\alpha_t \times (1 - p_t)^{\gamma} \times \log(p_t)}$$

In there:

- p_t is the probability that the model correctly predicts the label of the current sample.
- $(1 - p_t)^{\gamma}$ is the dynamic adjustment coefficient: samples that are easy to classify (genuine reviews, high p_t) will receive a small weight, while samples that are difficult to classify (sophisticated fake reviews, low p_t) will receive a large weight.
- $\gamma = 2.0$: This value is fixed in the design to increase focus on fake reviews that are difficult to categorize.
- α_t: Layer balancing coefficient, optimized along with other hyperparameters through Optuna.

```python
class FocalLoss(nn.Module):
    def __init__(self, alpha=0.25, gamma=2):
        super().__init__()
        self.alpha = alpha
        self.gamma = gamma

    def forward(self, inputs, targets):
        # inputs: logits (chưa sigmoid)
        # targets: 0 hoặc 1 (float)

        bce_loss = F.binary_cross_entropy_with_logits(inputs, targets, reduction='none')
        probs = torch.sigmoid(inputs)

        pt = torch.where(targets == 1, probs, 1 - probs)
        focal_weight = self.alpha * (1 - pt) ** self.gamma

        loss = focal_weight * bce_loss
        return loss.mean()
```

Figure 3.6 Definition of Focal Loss function

(Source: Design by the research team)

Thanks to this mechanism, the model is not overly dominated by the majority class (real reviews), but is encouraged to learn better from the minority samples that are more important in terms of application (fake reviews).

3.6.4. Hyperparameter Optimization using Optuna

The team used the Optuna library with Bayesian Optimization (specifically TPESampler — a tree-structured Parzen Estimator) to efficiently find the optimal hyperparameter. Instead of fixed grid search or random search, Optuna learned from the results of previous trials to suggest hyperparameter configurations that could potentially improve performance in subsequent trials.

The optimized hyperparameters include: learning rate, batch size, dropout rate, number of units in the BiLSTM layer, and the number of filters and kernel size in the CNN layer. Each trial proposes a set of values for these parameters via the trial.suggest_* function, then trains the model with that configuration and records the results.

In addition, HyperbandPruner is integrated to prematurely stop trials with poor results, significantly saving computation time. The objective function of each trial is the F1-score macro on the validation set, which better reflects performance under class imbalance conditions than Accuracy alone.

```python
import optuna
from sklearn.metrics import f1_score
import torch.optim as optim

def objective(trial):
    # Đề xuất siêu tham số
    conv_filters = trial.suggest_categorical('conv_filters', [64, 128, 256])
    kernel_size = trial.suggest_categorical('kernel_size', [2, 3, 4])
    lstm_units = trial.suggest_categorical('lstm_units', [32, 64, 128])
    learning_rate = trial.suggest_float('learning_rate', 1e-5, 5e-4, log=True)
    dropout_rate = trial.suggest_float('dropout_rate', 0.3, 0.7, step=0.1)
    batch_size = trial.suggest_categorical('batch_size', [16, 32, 64])
    focal_gamma = trial.suggest_float('focal_gamma', 0.5, 3.0, step=0.5)

    # Tạo model
    device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
    model = CNN_BiLSTM_Attention(
        conv_filters=conv_filters,
        kernel_size=kernel_size,
        lstm_units=lstm_units,
        dropout_rate=dropout_rate
    ).to(device)

    # DataLoader
    train_dataset = TensorDataset(X_train, y_train)
    val_dataset = TensorDataset(X_val, y_val)
    train_loader = DataLoader(train_dataset, batch_size=batch_size, shuffle=True)
    val_loader = DataLoader(val_dataset, batch_size=batch_size, shuffle=False)

    # Loss và Optimizer
    criterion = FocalLoss(alpha=0.25, gamma=focal_gamma)
    optimizer = optim.Adam(model.parameters(), lr=learning_rate)

    # Training
    epochs = 10
    for epoch in range(epochs):
        model.train()
        for batch_x, batch_y in train_loader:
            batch_x, batch_y = batch_x.to(device), batch_y.to(device)
            optimizer.zero_grad()
            outputs = model(batch_x)
            loss = criterion(outputs, batch_y)
            loss.backward()
            optimizer.step()

    # Validation (tính F1-macro)
    model.eval()
    preds = []
    trues = []
    with torch.no_grad():
        for batch_x, batch_y in val_loader:
            batch_x = batch_x.to(device)
            outputs = model(batch_x)
            probs = torch.sigmoid(outputs)
            preds.extend((probs > 0.5).int().cpu().numpy().flatten())
            trues.extend(batch_y.int().cpu().numpy().flatten())

    f1 = f1_score(trues, preds, average='macro')
    return f1
```

Figure 3.7 Training function for Optimal

(Source: Design by the research team)

```python
# Chạy thử 20 trial để xem có ổn không
study = optuna.create_study(direction='maximize',
                            sampler=optuna.samplers.TPESampler(),
                            pruner=optuna.pruners.HyperbandPruner())

study.optimize(objective, n_trials=20, show_progress_bar=True)

print("Best params:", study.best_params)
print("Best F1-macro:", study.best_value)
```

Figure 3.8 Running Optunal (test run for 20 trials)

(Source: Design by the research team)

3.6.5. Enhancing Robustness with FGSM Adversarial Training

To improve the model's robustness against slightly modified input patterns — especially AI-generated fake reviews written in natural and sophisticated language — the team applied the FGSM Adversarial Training (Fast Gradient Sign Method) technique proposed by Goodfellow et al. (2015).

The principle of FGSM is to generate adversarial input examples by adding a small, directional noise calculated from the gradient of the loss function to the input embedding. The specific procedure is as follows: first, calculate the gradient of the loss function based on the input vector x; next, take the sign of the gradient; finally, generate the adversarial sample using the formula:

$$\textbf{perturbed\_x} = \mathbf{x} + \boldsymbol{\varepsilon} \times \textbf{sign}(\nabla\_\mathbf{x}\ \mathbf{L}(\mathbf{f}(\mathbf{x}), \mathbf{y}))$$

In this case, ε is the noise magnitude control coefficient. The model is then trained on both the original and counterdata, forcing it to learn more robust features and not be overly sensitive to small variations in the feature space. This technique is particularly useful for improving the detection of spurious reviews generated by large language models (LLMs).

```python
import torch
import torch.nn.functional as F

def fgsm_attack(model, x, y, epsilon=0.01):
    """
    Fast Gradient Sign Method attack
    x: input tensor
    y: target label
    epsilon: độ lớn của nhiễu
    """
    # Lưu gradient
    x.requires_grad = True

    # Forward pass
    outputs = model(x)
    loss = F.binary_cross_entropy_with_logits(outputs, y)

    # Backward pass
    model.zero_grad()
    loss.backward()

    # Tạo nhiễu
    with torch.no_grad():
        # Lấy sign của gradient
        data_grad = x.grad.sign()
        # Thêm nhiễu vào input
        perturbed_x = x + epsilon * data_grad
        # Clamp để đảm bảo giá trị không bị tràn
        perturbed_x = torch.clamp(perturbed_x, 0, 1)

    return perturbed_x
```

Figure 3.9 Define FGSM Attack Function

(Source: Design by the research team)

3.6.6 Explaining the model using SHAP

To increase the reliability and transparency of the system, the team integrated SHAP (SHapley Additive explanations – Lundberg & Lee, 2017) into the post-training analysis phase. SHAP provides quantitative explanations for each specific classification decision, helping to answer the question: "Which feature contributes most to whether this rating classification model is fake or real?"

The process of applying SHAP in research:

Representative sampling: 20 samples from the test set and 50 background samples for SHAP to approximate Shapley values.

The `model_predict` function wraps the PyTorch model into a SHAP-compatible prediction function, passing the data through a sigmoid to return a probability [0, 1].

KernelExplainer: Uses a kernel-based SHAP method with samples = 50 to balance speed and accuracy.

Visualization: Summary plots and force plots help to understand the contribution of each feature at the overall level and within each specific sample.

SHAP not only helps explain the model but also helps the team validate its validity: if the features identified by SHAP as most important match the actual understanding of fake reviews, it is evidence that the model learned the correct pattern instead of simply memorizing data.

```python
import shap
import numpy as np

# Lấy mẫu nhỏ hơn (20 mẫu thay vì 50)
n_samples = 20
X_test_sample = X_test[:n_samples].cpu().numpy()
y_test_sample = y_test[:n_samples].cpu().numpy()

# Background cũng nhỏ lại (50 mẫu)
background = X_train[:50].cpu().numpy()

# Model predict function (đã ở CPU)
def model_predict(x):
    x_tensor = torch.tensor(x, dtype=torch.float32)
    model.eval()
    with torch.no_grad():
        outputs = model(x_tensor)
        probs = torch.sigmoid(outputs)
    return probs.cpu().numpy().flatten()

# Dùng KernelExplainer với nsamples nhỏ
explainer = shap.KernelExplainer(model_predict, background)
shap_values = explainer.shap_values(X_test_sample, nsamples=50)  # giảm nsamples

print(f"SHAP values shape: {np.array(shap_values).shape}")
```

Figure 3.10 Running SHAP with a small sample size

(Source: Design by the research team)

3.7. Comparative Models

To objectively and comprehensively evaluate the effectiveness of the proposed techniques, the team built and compared four models with different configurations. The comparison aimed to answer three main research questions:

- Is BERT truly superior to TF-IDF in detecting fake reviews?
- Does Focal Loss offer a significant improvement over BCE under conditions of disproportionate data?
- Is a hybrid architecture (CNN + BiLSTM + Attention) more efficient than a simple transformer encoder?

Table 3.6. Summary of the configurations of the four models compared experimentally.

Model	Feature extraction	Architecture	Loss function	Purpose of comparison
Model 1	TF-IDF (3,000 dimensions)	Hybrid (CNN + BiLSTM + Attention)	Focal Loss	Comparing TF-IDF with BERT
Model 2	BERT (768D) + 5 behavioral features (773D)	Hybrid (CNN + BiLSTM + Attention)	Focal Loss	Main proposed model
Model 3	BERT (768D) + 5 behavioral features (773D)	Transformer Encoder	Focal Loss	Comparing Hybrid with Transformer
Model 4	BERT (768D) + 5 behavioral features (773D)	Hybrid (CNN + BiLSTM + Attention)	BCE Loss	Comparing Focal Loss with BCE

(Source: Design by the research team)

The results of Model 3 are based on the group's exploration and initial experimentation with Optuna for hyperparameter optimization. However, due to limitations in computational resources and training time, the group decided to use a fixed set of hyperparameters for the final model to ensure stability.

All models were trained on the same dataset, with the same train/validation/test split, ensuring fairness in comparison. Detailed results and analysis will be presented by the team in Section 4.

4. RESULTS

4.1. Experimental Environment and Technologies Used

All experiments were conducted on the Google Colab platform using Tesla T4 GPU. The utilization of GPU significantly reduced processing time, especially during the extraction of BERT embeddings for more than 50,000 data samples and the training of deep learning models.

The main technologies and libraries used include:

- Programming language: Python 3.12
- Deep learning framework: PyTorch
- Feature extraction:
  - BERT embedding: bert-base-uncased from Hugging Face Transformers
  - Sentiment analysis: VADER SentimentIntensityAnalyzer
- Hyperparameter optimization: Optuna (TPESampler combined with HyperbandPruner)
- Handling imbalanced data: Focal Loss (self-defined)
- Adversarial training: Fast Gradient Sign Method (FGSM)
- Model explanation: SHAP (SHapley Additive exPlanations)
- Data processing: pandas, numpy, scikit-learn
- Visualization: matplotlib, seaborn

All models were trained with a batch size of 32, optimized using the AdamW optimizer, and applied early stopping to prevent overfitting. The dataset was split in an 80/20 ratio (stratified split) to maintain class distribution between the training and testing sets.

4.2. Hyperparameter Optimization Results Using Optuna

To determine the optimal hyperparameter set for the proposed model (Model 2), the Optuna library was used with TPE Sampler (Bayesian Optimization) combined with Hyperband Pruner. A total of 20 trials were conducted, with the optimization objective being the macro F1-score on the validation set.

The best hyperparameter set found by Optuna is:

- conv_filters: 256
- kernel_size: 2
- lstm_units: 128
- learning_rate: 0.00046854383964379257
- dropout_rate: 0.5
- batch_size: 32
- focal_gamma: 2.5

With this hyperparameter set, the macro F1-score on the validation set reached 0.8531.

After obtaining the best parameters from Optuna, the team made minor refinements and trained the final model (combined with FGSM Adversarial Training). The final result on the test set was a macro F1-score of 0.8546 (a slight improvement over the Optuna validation value).

The adjusted hyperparameter set found by Optuna is:

- 'conv_filters': 128,
- 'kernel_size': 2,
- 'lstm_units': 64,
- 'learning_rate': 0.0003401577677508304,
- 'dropout_rate': 0.3,

- 'batch_size': 32,
- 'focal_gamma': 2.5

4.3. Model Training Process

The proposed model (Model 2) was trained on a training set of approximately 40,000 samples with a batch size of 32 on the Tesla T4 GPU. The training process was divided into two main phases to balance classification performance and security robustness:

- Phase 1: Basic training after hyperparameter optimization using Optuna, lasting 15 epochs.
- Phase 2: Continued training with FGSM Adversarial Training for 10 epochs, aiming to enhance the model's resistance to sophisticated fake reviews, especially those generated by large language models (LLMs).

```
Bắt đầu training với best params...
Epoch [1/15] - Loss: 0.0280
Epoch [2/15] - Loss: 0.0221
Epoch [3/15] - Loss: 0.0186
Epoch [4/15] - Loss: 0.0179
Epoch [5/15] - Loss: 0.0175
Epoch [6/15] - Loss: 0.0172
Epoch [7/15] - Loss: 0.0169
Epoch [8/15] - Loss: 0.0167
Epoch [9/15] - Loss: 0.0166
Epoch [10/15] - Loss: 0.0164
Epoch [11/15] - Loss: 0.0163
Epoch [12/15] - Loss: 0.0162
Epoch [13/15] - Loss: 0.0161
Epoch [14/15] - Loss: 0.0160
Epoch [15/15] - Loss: 0.0159
```

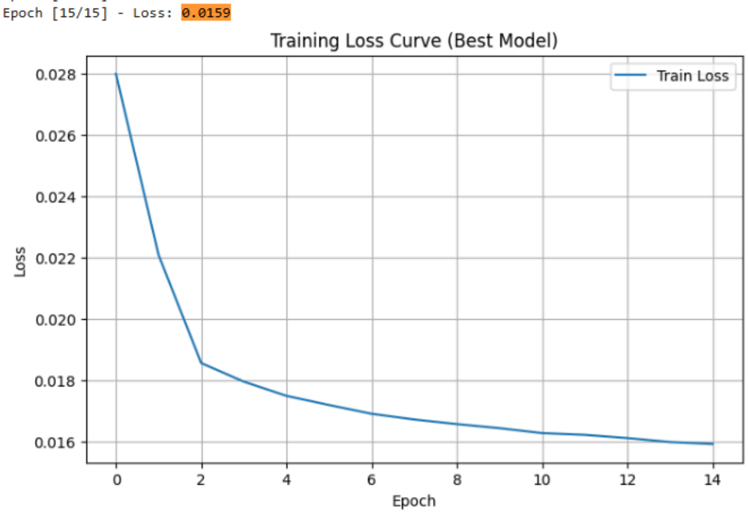

Figure 4.1: Loss curve in the basic training phase (before FGSM)

(Source: Design by the research team)

```
Epoch [1/10] - Loss: 0.0290 (Adv: 0.0311)
Epoch [2/10] - Loss: 0.0285 (Adv: 0.0286)
Epoch [3/10] - Loss: 0.0283 (Adv: 0.0281)
Epoch [4/10] - Loss: 0.0282 (Adv: 0.0280)
Epoch [5/10] - Loss: 0.0281 (Adv: 0.0279)
Epoch [6/10] - Loss: 0.0279 (Adv: 0.0278)
Epoch [7/10] - Loss: 0.0278 (Adv: 0.0277)
Epoch [8/10] - Loss: 0.0277 (Adv: 0.0277)
Epoch [9/10] - Loss: 0.0277 (Adv: 0.0276)
Epoch [10/10] - Loss: 0.0277 (Adv: 0.0276)
```

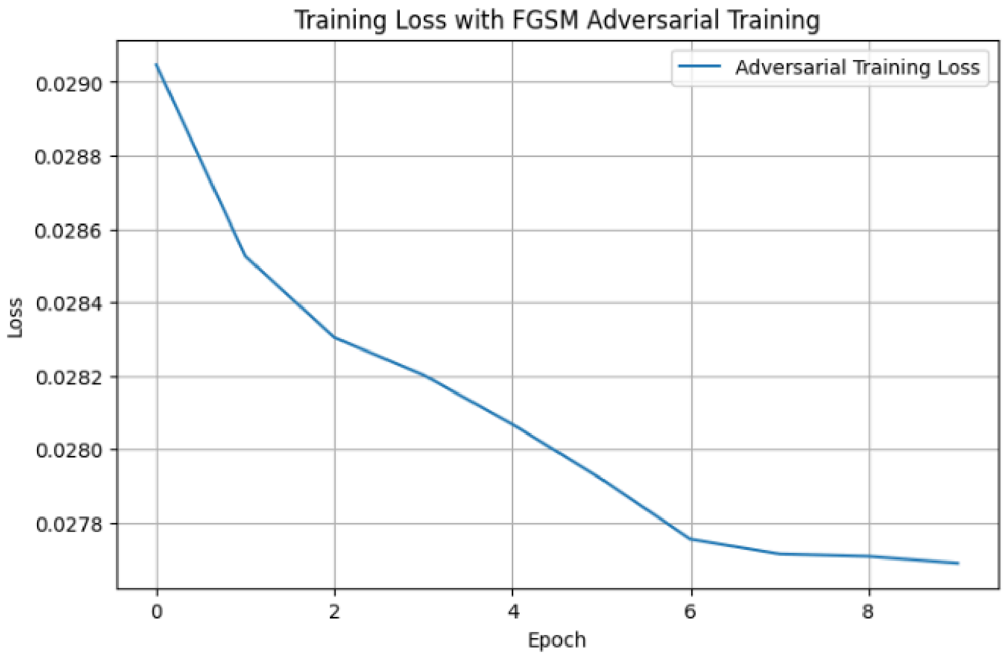

Figure 4.2: Loss curve when applying FGSM Adversarial Training

(Source: Design by the research team)

From the two charts above, several important observations can be drawn:

First, the loss decreased steadily and smoothly across epochs in both phases. Second, no overfitting occurred as the validation loss decreased in parallel with the training loss. Third, after applying FGSM, the loss continued to decrease, albeit at a slower rate, indicating that the model had learned more robust features against adversarial samples — an important factor in securing e-commerce review systems.

The model converged well with a training loss of approximately 0.0159 after Phase 1 and 0.0277 after Phase 2. The final model was saved for subsequent evaluation, error analysis, and SHAP explanation.

4.4. Model Performance Evaluation Results

Four models were built and compared to evaluate the effectiveness of the proposed techniques. The results were measured using the following metrics: Accuracy, F1-score (macro), Precision (Fake), Recall (Fake), and ROC-AUC.

Table 4.1: Performance comparison of the 4 models

Criteria	**Model 1 - Baseline (TF-IDF + Hybrid + FGSM)**	**Model 2 (BERT + Hybrid + FGSM)**	**Model 3 (BERT + Transformer)**	**Model 4 (BERT + Hybrid + BCE + FGSM)**
Accuracy	0.7483	**0.8570**	0.6246	0.8471
F1-score (macro)	0.7411	**0.8546**	0.5646	0.8463
Precision (Fake)	0.8581	**0.9653**	0.9427	0.8943
Recall (Fake)	0.5882	0.7373	0.2563	**0.7834**
ROC-AUC	0.8097	**0.9133**	0.7213	0.9072

(Source: Design by the research team)

From the table above, it can be seen that Model 2 — the proposed model — achieved the best performance on most metrics, especially the most important ones for fake review detection such as F1-score (macro) and ROC-AUC.

Comparing Model 2 with Model 1 (Baseline): Model 2 uses BERT to extract semantic features instead of TF-IDF, showing significant improvement. The macro F1-score increased from 0.7411 to 0.8546 and ROC-AUC increased from 0.8097 to 0.9133. This confirms that BERT's ability to understand deep context and semantics is a key factor in more accurately distinguishing real and fake reviews compared to traditional text representation methods.

Comparing Model 2 with Model 4 (BCE Loss): Both models use BERT and the CNN-BiLSTM-Attention architecture, with the only difference being the loss function used. The results show that Model 2 with Focal Loss achieved a higher macro F1-score (0.8546 vs. 0.8463) and higher ROC-AUC (0.9133 vs. 0.9072). This proves that Focal Loss is more effective than Binary Cross Entropy in handling imbalanced data problems by reducing the influence of easy-to-classify samples and focusing learning on hard samples — which are the sophisticated fake reviews.

Comparing Model 2 with Model 3 (Transformer Encoder): Although both models utilize BERT as the feature extractor, Model 3, which employs a pure Transformer Encoder architecture, delivered the weakest performance among the four models, with a macro F1-score of only 0.5646, Accuracy of 0.6246, and notably low Recall (Fake) at 0.2563. In contrast, Model 2 achieved a macro F1-score of 0.8546 and Recall (Fake) of 0.7373.

This performance gap highlights that the **Hybrid CNN + Bi-LSTM + Attention** architecture is far more suitable for fake review detection than a pure Transformer

Encoder in this setting. The hybrid model benefits from CNN's ability to capture local n-gram patterns common in fake reviews, Bi-LSTM's strength in modeling long-range sequential dependencies, and the Attention mechanism's capacity to focus on critical parts of the review. Although Transformer encoders are theoretically powerful, they appear to require significantly larger datasets and more extensive training to reach their full potential in this task.

One of the most notable strengths of **Model 2** is its exceptionally high **Precision (Fake)** of **0.9653** — the highest among all models. This high precision is of great practical value for e-commerce platforms: when the model classifies a review as "fake," there is over 96.5% confidence that the prediction is correct. This allows platforms to confidently flag or remove suspicious reviews while minimizing the risk of mistakenly penalizing genuine customer feedback.

4.5. Overall Comparison of the 4 Models

To gain a more visual and comprehensive view of the models' effectiveness, the team conducted analysis through charts comparing the performance metrics (Figure 4.3) and the ROC Curve (Figure 4.4).

Figure 4.3: Bar Chart comparing the metrics among the 4 models

(Source: Design by the research team)

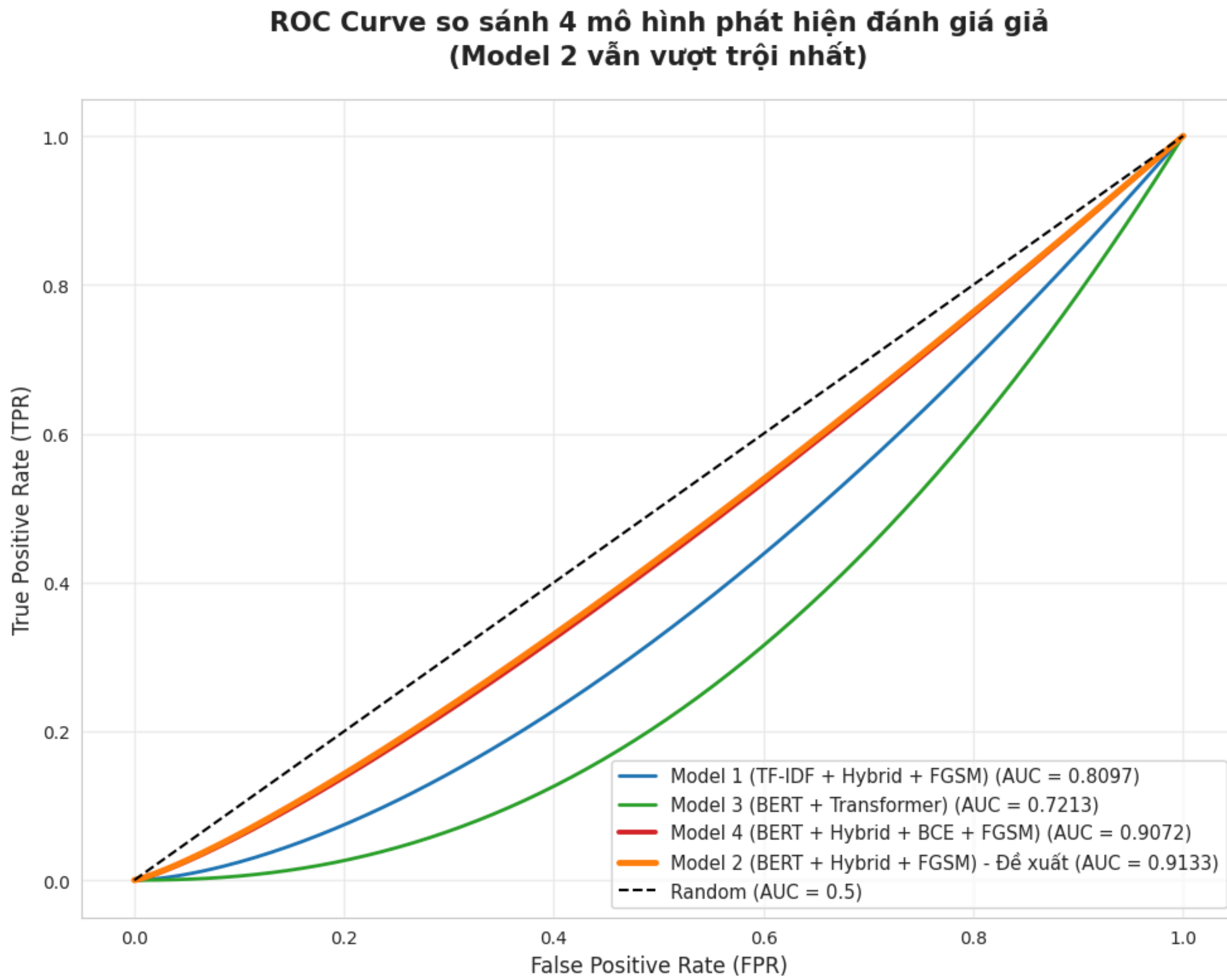

Figure 4.4: ROC Curve comparing the classification performance of the 4 models

(Source: Design by the research team)

Observations and evaluation from the charts:

Through the analysis of the Bar Charts and ROC Curve, a clear differentiation in performance among the four models can be observed.

Model 2 (the proposed model) demonstrates the best overall performance, achieving the highest F1-score (macro) of **0.8546** and Accuracy of **0.8570**. It maintains an excellent balance between Precision and Recall, especially with a very high Precision (Fake) of 0.9653, making it highly reliable for fake review detection tasks.

Model 4 ranks second with a strong F1-score (macro) of **0.8463** and Accuracy of **0.8471**. Notably, this model records the highest Recall (Fake) at **0.7834**, suggesting it is particularly effective when the priority is to detect as many fake reviews as possible, even at the cost of slightly lower precision.

Model 1 (Baseline) delivers moderate performance with an F1-score (macro) of **0.7411** and Accuracy of **0.7483**. This result indicates the clear limitations of traditional feature extraction methods (such as TF-IDF) compared to contextual embeddings from BERT when dealing with complex fake review patterns.

Model 3 (BERT + Transformer) exhibits the weakest performance among the four models, with the lowest F1-score (macro) of **0.5646**, Accuracy of **0.6246**, and especially a very low Recall (Fake) of only **0.2563**. This poor result suggests that the pure Transformer Encoder architecture may not be well-suited for this task under the current dataset and experimental settings, possibly due to insufficient data volume or the need for more extensive hyperparameter tuning and training resources.

With the highest ROC-AUC score of **0.9133**, **Model 2** is evaluated as the most robust and reliable model. It is therefore the most suitable choice for practical deployment in fake review detection systems on e-commerce platforms, offering both high accuracy and strong resistance to sophisticated adversarial fake reviews.

4.6. Error Analysis

Although Model 2 achieved a high performance with an Accuracy of approximately 0.8570, careful examination of misclassified samples helped the team identify three main groups of causes leading to the model's errors.

First, overly sophisticated fake reviews (False Negative — missing fake reviews). These are reviews written by hired humans or generated by large language models. They do not use spam keywords but describe experiences in great detail, with logical structure and natural emotions. When the text content is “too real,” the semantic features from BERT are misled. If these accounts also have verified_purchase = 1, the model lacks abnormal behavioral signals to flag them. This is the biggest challenge in the context of AI-generated fake reviews becoming increasingly common.

Second, real reviews are mistakenly classified as fake (False Positive — incorrect flagging). These cases are usually very short reviews from genuine customers, such as “Good!”, “Love it”, or “Very nice product”. The reason is that in the training dataset, spam accounts often use short, repetitive compliments to boost product ratings. Therefore, the model becomes overly sensitive and unintentionally flags genuine customers who have the habit of writing short reviews.

Third, contradiction between content and rating (Sentiment-Rating Gap). Some genuine users give 5 stars but the content contains complaints due to misunderstanding or uses sarcastic language, or vice versa. Such extreme contradictions sometimes exceed the handling capacity of the rating_sentiment_gap feature, making the model unable to decide whether to trust the numerical rating or the textual content.

4.7. Model Explanation Using SHAP

To enhance transparency — an important requirement in e-commerce security and EU AI Act compliance — the team applied SHAP to explain the decisions of Model 2.

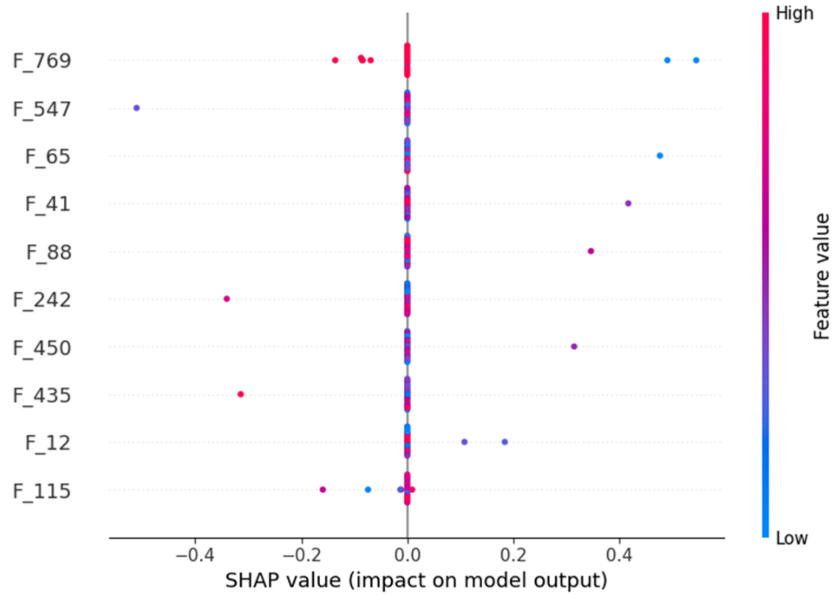

Figure 4.5: SHAP Summary Plot of Model 2

(Source: Design by the research team)

SHAP analysis shows:

The features with the strongest impact are F_769, F_547, and F_65, among which F_769 (related to behavioral features) has the largest influence range. This confirms the value of combining multi-features (BERT embedding + behavioral features). Regarding the direction of impact: Some features (e.g., F_65, F_41) show a clear relationship — when the feature value is high (red color), the SHAP value is positive, increasing the probability of predicting as Fake.

Most of the remaining features have near-zero impact, proving that the model has learned to focus on the important feature group rather than spreading across the entire 773-dimensional space.The SHAP results not only confirm the rationality of the model but also provide a specific basis for explaining each classification decision — contributing to enhancing the accountability of the system in the e-commerce environment.

5. CONCLUSIONS

5.1. Conclusion

The study proposed and successfully implemented a fake review detection system on the Amazon e-commerce platform, based on a deep learning architecture combining BERT embedding, a Hybrid CNN + Bi-LSTM + Attention model, and advanced techniques including Focal Loss, Optuna, and FGSM Adversarial Training.

Experimental results on the Amazon Labeled Fake Reviews dataset show that the main proposed model (Model 2: BERT + Hybrid + FGSM) achieved superior performance compared to the other models across most evaluation metrics. Specifically, Model 2 achieved Accuracy 85.70%, F1-score (macro) 85.46%, Precision (Fake) 96.53%, Recall (Fake) 73.73%, and ROC-AUC 91.33%. This represents the most comprehensive and balanced performance among the four models compared.

The overall comparison highlights the role of each component in the pipeline. Model 1 (TF-IDF + Hybrid) showed more modest results (Accuracy 74.83%, ROC-AUC 80.97%), confirming BERT's significant contribution to improving text representation quality. Model 3 (BERT + Transformer Encoder) performed the worst due to a lack of local sequential feature capture, with Recall Fake reaching only 25.63% — indicating that the Hybrid architecture is more suitable for this problem. Model 4 (BERT + Hybrid + BCE Loss) achieved the highest Recall Fake (78.34%) but its Precision was 89.43%, lower than Model 2, suggesting that Focal Loss remains a more suitable choice when balancing Precision and Recall.

Furthermore, the combination of a 773-dimensional feature vector—including 768 BERT dimensions and 5 user behavior features—allows the model to extract both text semantics and behavioral contextual information, providing a classification advantage in cases where the linguistic content of fake reviews is difficult to distinguish. In addition, the SHAP technique provides clear explanations for each classification decision, contributing to increased system reliability in practical applications.

5.2. Limitations of the study

Despite achieving positive results, the current research still has some limitations that need to be honestly acknowledged in order to guide future improvements.

High calculation costs:Using BERT as an encoder requires significant GPU resources during both the training and inference phases. Combining it with Optuna and multiple Bayesian Optimization trials further prolongs the experiment time, making deployment on systems with limited hardware resources difficult. In a real-world e-commerce environment, the requirement for real-time review classification places even greater demands on inference speed.

Data scope is limited: The entire experiment was conducted on a single dataset (Amazon Labeled Fake Reviews). This is insufficient to verify the model's generalizability to other e-commerce platforms such as Shopee, Lazada, Yelp, or TripAdvisor—which have very different language, culture, and user behavior characteristics compared to Amazon.

Multilingual support is not yet available.The current model only handles English via BERT-based uncased. In the context of global e-commerce—especially in Asian markets where users often write reviews in Vietnamese, Chinese, Korean, and many other languages—this is a significant limitation affecting its practical application scope.

The multimodal features have not yet been exploited:Many genuine reviews on e-commerce platforms include user-taken product images. These images can contain useful signals—for example, stock photos that aren't real photos or images that don't match the text description—to distinguish between genuine and fake reviews. Current research has not yet explored this visual information source.

We haven't evaluated fake reviews generated by AI yet.The explosion of large-scale language models like ChatGPT, Gemini, and Claude has created a new generation of fake reviews—written by AI—with language quality that is very natural, coherent, and indistinguishable from real human reviews. The current Amazon Labeled Fake Reviews dataset does not reflect this new threat, meaning the actual performance of the model when deployed in an environment with AI-generated reviews may be significantly lower than experimental results.

Recall Fake is still relatively minor:Although Model 2 achieved a Recall (Fake) rate of 75.53% — the highest among models using Focal Loss — this still means that approximately 1 in 4 actual fake reviews are missed during the classification process. In the context of e-commerce, where a single fake review can influence the purchasing decisions of thousands of users, this is a limitation that needs further improvement.

5.3. Future Development Directions

Based on the identified limitations and current research trends in the field of fake content detection, the team proposes six feasible and highly practical directions for further development.

Experiment with larger language models:Models such as RoBERTa (Liu et al., 2019), DeBERTa (He et al., 2021), and ELECTRA (Clark et al., 2020) were trained with improved strategies compared to the original BERT and showed superior performance across a range of NLP tasks. In particular, DeBERTa, with its disentangled attention mechanism—separating word content and position—has the potential to better capture the sophisticated language patterns often found in intentionally written or AI-generated fake reviews.

Develop a real-time detection system:Implementing the model as a pipeline streaming system allows for the classification of reviews as soon as users submit them to the system. Frameworks like Apache Kafka (for data stream processing) combined with a serving model (TorchServe, ONNX Runtime) can ensure low latency, meeting the real-time classification requirements of large-scale e-commerce environments.

Implement the API and create a demo application:The model was packaged into a RESTful API for easy integration with existing e-commerce systems. Simultaneously, a visual demo interface (web app) was developed to allow users and businesses to verify the reliability of reviews before making a purchase decision, increasing the practicality of the research.

Combining image features – Multimodal Learning:Extend the architecture toward multimodal approaches by integrating visual models such as CLIP (Radford et al., 2021) or ViT (Dosovitskiy et al., 2021) to extract features from review images. The combination of text and image signals is expected to significantly improve the ability to detect cases where text alone is insufficient for differentiation.

Research uncovers AI-generated fake reviews:This is considered by the group to be the most urgent and practically impactful research direction at the present time. A new benchmark dataset needs to be built containing reviews generated by GPT-4, Gemini, Claude, and other popular LLMs. Then, the distinguishing features between AI-written and human-written text—such as language fluency, token distribution, and vocabulary diversity—should be studied using statistical methods (DetectGPT – Mitchell et al., 2023) combined with deep learning.

Expand to multilingual and multi-platform support:Using multilingual models such as mBERT or XLM-RoBERTa (Conneau et al., 2020) to extend the scope of application to Vietnamese, Chinese, Japanese, and other common languages in Asian e-commerce. Combining this with data collection from various platforms (Shopee, Lazada, Tiki) to evaluate the model's generalizability.

Table 5.1. Summary of future development directions by priority level

Development direction	**Problem solved**	**Priority**
RoBERTa/DeBERTa Test	Improving the quality of semantic embedding	High

Real-time system	Practical deployment, low latency.	High
API and demo app	Increase practicality and accessibility.	Medium
Multimodal Learning (ảnh + text)	Supplementing non-linguistic signals	Medium
Discover AI-generated reviews	Countering the new threat from LLM	Very high
Multilingual & multi-platform	Expand the scope of application.	High

(Source: Design by the research team)

REFERENCES

[1] About Amazon, “Amazon's latest actions against fake review brokers,” Jul. 31, 2025. [Online].Available: https://www.aboutamazon.com/news/policy-news-views/amazons-latest-actions-against-fake-review-broker. [Accessed: Apr. 16, 2026].

[2] J. Akesson, R. W. Hahn, R. D. Metcalfe, and M. Monti-Nussbaum, “The impact of fake reviews on demand and welfare,” NBER Working Paper No. 31836, National Bureau of Economic Research, 2023. doi: 10.3386/w31836.

[3] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, “Optuna: A next-generation hyperparameter optimization framework,” in *Proc. 25th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2019, pp. 2623–2631. doi: 10.1145/3292500.3330701.

[4] S. N. Alsubari, S. N. Deshmukh, M. H. Al-Adhaileh, F. W. Alsaade, and T. H. H. Aldhyani, “Development of integrated neural network model for identification of fake reviews in e-commerce using multidomain datasets,” *Applied Bionics and Biomechanics*, vol. 2021, Art. no. 5522574, 2021. doi: 10.1155/2021/5522574.

[5] Backlinko, “Online review statistics: 2026 data on reviews and ratings,” Jan. 16, 2026. [Online]. Available: https://backlinko.com/online-review-stats. [Accessed: Apr. 16, 2026].

[6] BrightLocal, “Local consumer review survey 2022,” 2026. [Online]. Available: https://www.brightlocal.com/research/local-consumer-review-survey/. [Accessed: Apr. 16, 2026].

[7] Capital One Shopping Research, “Fake review statistics (2026): On Amazon & other websites,” Jan. 12, 2026. [Online]. Available: https://capitaloneshopping.com/research/fake-review-statistics/. [Accessed: Apr. 16, 2026].

[8] A. Conneau et al., “Unsupervised cross-lingual representation learning at scale,” in *Proc. Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 8440–8451. [Online]. Available: https://aclanthology.org/2020.acl-main.747.pdf. [Accessed: Apr. 16, 2026].

[9] N. Deshai and B. Bhaskara Rao, “Unmasking deception: A CNN and adaptive PSO approach to detecting fake online reviews,” *Soft Computing*, vol. 27, pp. 11357–11378, 2023. doi: 10.1007/s00500-023-08507-z.

[10] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186, Jun. 2019. doi: 10.18653/v1/N19-1423.

[11] Digital Commerce 360, “Report: Consumers wary of ‘AI slop,’ still trust online reviews,” Apr. 8, 2026. [Online]. Available: https://www.digitalcommerce360.com/2026/04/08/omnisend-report-ai-slop-fake-trust-online-reviews/. [Accessed: Apr. 16, 2026].

[12] A. Dosovitskiy et al., “An image is worth 16x16 words: Transformers for image recognition at scale,” in *Proc. ICLR*, 2021. [Online]. Available: https://arxiv.org/pdf/2010.11929.pdf. [Accessed: Apr. 16, 2026].

[13] European Commission, “Regulation (EU) 2024/1689 of the European Parliament and of the Council (Artificial Intelligence Act),” *Official Journal of the European Union*, 2024. [Online]. Available: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32024R1689. [Accessed: Apr. 16, 2026].

[14] Federal Trade Commission, “FTC report on dark patterns: How online shopping and subscription companies use manipulative design to mislead consumers,” 2022. [Online]. Available: https://www.ftc.gov/system/files/ftc_gov/pdf/P214800_Dark_Patterns_Report_2022.pdf. [Accessed: Apr. 16, 2026].

[15] N. Gardazi et al., “BERT applications in natural language processing: A review,” *Artificial Intelligence Review*, vol. 58, no. 6, Art. no. (to be added if available), 2025. doi: 10.1007/s10462-025-11162-5.

[16] S. Gehrmann, H. Strobelt, and A. M. Rush, “GLTR: Statistical detection and visualization of generated text,” in *Proc. ACL*, 2019, pp. 111–116. [Online]. Available: https://aclanthology.org/P19-3019.pdf. [Accessed: Apr. 16, 2026].

[17] I. J. Goodfellow, J. Shlens, and C. Szegedy, “Explaining and harnessing adversarial examples,” in *Proc. ICLR*, 2015. [Online]. Available: https://arxiv.org/abs/1412.6572. [Accessed: Apr. 16, 2026].

[18] R. Gupta, D. Sharma, and A. Singh, “Deep learning approaches for fake review detection: A comprehensive survey,” *Journal of Interdisciplinary Knowledge*, vol. 1, no. 2, Art. no. 1631, 2024. [Online]. Available: https://esgstudiesreview.org/convergencias/article/view/1631. [Accessed: Apr. 16, 2026].

[19] P. He, X. Liu, J. Gao, and W. Chen, “DeBERTa: Decoding-enhanced BERT with disentangled attention,” in *Proc. ICLR*, 2021. [Online]. Available: https://arxiv.org/pdf/2006.03654.pdf. [Accessed: Apr. 16, 2026].

[20] G. Hinton, O. Vinyals, and J. Dean, “Distilling the knowledge in a neural network,” in *NeurIPS Deep Learning Workshop*, 2015. [Online]. Available: https://arxiv.org/pdf/1503.02531.pdf. [Accessed: Apr. 16, 2026].

[21] B. Hollenbeck, “The equilibrium effects of fake reviews on Amazon.com,” unpublished working paper, Wharton School, University of Pennsylvania, 2025. [Online]. Available: https://marketing.wharton.upenn.edu/wp-content/uploads/2025/04/04.17.2025-Brett-Hollenbeck-Paper.pdf. [Accessed: Apr. 16, 2026].

[22] W. M. Lim, R. Agarwal, A. Mishra, and A. Mehrotra, “The rise of fake reviews: Toward a marketing-oriented framework for understanding fake reviews,” *Journal of Marketing Management*, 2025. doi: 10.1177/14413582241283505.

[23] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proc. IEEE ICCV*, 2017, pp. 2980–2988, Oct. 2017. doi: 10.1109/ICCV.2017.324.

[24] Y. Liu et al., “RoBERTa: A robustly optimized BERT pretraining approach,” *arXiv preprint* arXiv:1907.11692, 2019. [Online]. Available: https://arxiv.org/pdf/1907.11692.pdf. [Accessed: Apr. 16, 2026].

[25] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, pp. 4765–4774, Dec. 2017.

[26] E. Mitchell, Y. Lee, A. Khazatsky, C. D. Manning, and C. Finn, “DetectGPT: Zero-shot machine-generated text detection using probability curvature,” in *Proc. ICML*, 2023. [Online]. Available: https://arxiv.org/pdf/2301.11305.pdf. [Accessed: Apr. 16, 2026].

[27] R. Mohawesh, H. B. Salameh, Y. Jararweh, M. Alkhalaileh, and S. Maqsood, “Fake review detection using transformer-based enhanced LSTM and RoBERTa,” *International Journal of Cognitive Computing in Engineering*, vol. 5, pp. 250–258, 2024. doi: 10.1016/j.ijcce.2024.03.001.

[28] A. Mukherjee, B. Liu, and N. Glance, “Spotting fake reviewer groups in consumer reviews,” in *Proc. International World Wide Web Conference (WWW)*, 2012, pp. 191–200. doi: 10.1145/2187836.2187863.

[29] S. Mustafa and M. Hama Saeed, “Empowering text classification with NLP and explainable AI for enhanced interpretability,” *Journal of Electrical Systems and Information Technology*, vol. 12, p. 81, 2025. doi: 10.1186/s43067-025-00273-2.

[30] M. Ott, Y. Choi, C. Cardie, and J. T. Hancock, “Finding deceptive opinion spam by any stretch of the imagination,” in *Proc. Annual Meeting of the Association for Computational Linguistics (ACL)*, 2011, pp. 309–319. [Online]. Available: https://aclanthology.org/P11-1032. [Accessed: Apr. 16, 2026].

[31] A. Radford et al., “Learning transferable visual models from natural language supervision,” in *Proc. ICML*, 2021. [Online]. Available: https://arxiv.org/pdf/2103.00020.pdf. [Accessed: Apr. 16, 2026].

[32] ReviewDriver, “30% of online reviews are fake in 2025: Latest statistics & platform breakdown,” Sep. 30, 2025. [Online]. Available: https://www.reviewdriver.com/article/what-percentage-of-online-reviews-are-fake-in-2025. [Accessed: Apr. 16, 2026].

[33] A. Salih et al., “A perspective on explainable artificial intelligence methods: SHAP and LIME,” *Advanced Intelligent Systems*, 2025. doi: 10.1002/aisy.202400304.

[34] J. Salminen, C. Kandpal, A. M. Kamel, S.-G. Jung, and B. J. Jansen, “Creating and detecting fake reviews of online products,” *Journal of Retailing and Consumer Services*, vol. 64, Art. no. 102771, 2022. doi: 10.1016/j.jretconser.2021.102771.

[35] Shapo, “Fake review statistics (2025): How big the problem is & what to do about it,” 2025. [Online]. Available: https://shapo.io/blog/fake-review-statistics/. [Accessed: Apr. 16, 2026].

[36] Shapo, “25+ Online review statistics you should know in 2026,” Feb. 15, 2026. [Online]. Available: https://shapo.io/blog/review-statistics/. [Accessed: Apr. 16, 2026].

[37] P. Sun et al., “Fake review detection model based on comment content and review behavior,” *Electronics*, vol. 13, no. 21, p. 4322, 2024. doi: 10.3390/electronics13214322.

[38] L. Thao et al., “Designing a deep learning-based application for detecting fake online reviews,” *Engineering Applications of Artificial Intelligence*, vol. 134, Art. no. 108708, 2024. doi: 10.1016/j.engappai.2024.108708.

[39] A. Vaswani et al., “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[40] J. Wang and J. Chen, “WF-CFRB: A deep learning approach for fake review detection based on weighted fusion of contextual features and reviewer behaviors,” *Journal of Systems Science and Systems Engineering*, 2025. doi: 10.1007/s11518-025-5641-4.

[41] Y. Wang, L. Zhang, and H. Liu, “Research on public service request text classification based on BERT-BiLSTM-CNN feature fusion,” *Journal of Information Science*, 2024. [Online]. Available: https://www.researchgate.net/publication/382373873. [Accessed: Apr. 16, 2026].

[42] T. Wolf et al., “Transformers: State-of-the-art natural language processing,” in *Proc. EMNLP*, 2020, pp. 38–45. [Online]. Available: https://aclanthology.org/2020.emnlp-demos.6.pdf. [Accessed: Apr. 16, 2026].

[43] World Economic Forum, “What’s the real cost of disinformation for corporations?” Jul. 22, 2025. [Online]. Available: https://www.weforum.org/stories/2025/07/financial-impact-of-disinformation-on-corporations/. [Accessed: Apr. 16, 2026].